THE INVERSE IMAGE OF A CONSTRUCTIBLE SUBSET ALONG ANY MORPHISM IS CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-inverse-image-constructible is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 4339-4345 and proof span 4347-4366 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof ends with the author's phrase 'we win'; it is preserved verbatim.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The reduction of constructibility to retrocompact opens, cited from the topology chapter: it suffices to show that inverse images of retrocompact opens are retrocompact. Cited at proof lines 4564-4567. Reduces the constructible case to a statement about retrocompact opens.
- An open immersion with retrocompact source is quasi-compact, and quasi-compactness survives base change, cited from the schemes chapter. Cited at proof lines 4568-4571. Shows that the base change of the retrocompact open is quasi-compact, which is exactly retrocompactness in the source.
- The characterization of locally constructible subsets by their traces on affine opens, cited from the properties chapter. Cited at proof lines 4578-4580. Reduces the locally constructible case to the constructible case on an affine chart.

Source reference. Original lemma 054C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6635–6635. Same-chapter reference used in the retained author text at source line(s) 4362, 4415, 4433. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01K5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2038–2038. Same-chapter reference used in the retained author text at source line(s) 1319, 2335, 2748, 3954, 4177, 4356. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 005I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 210–210. Same-chapter reference used in the retained author text at source line(s) 4351. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $E \subset Y$ be a subset. If E is (locally) constructible in Y, then $f^{-1}(E)$ is (locally) constructible in X.*

Proof. To show that the inverse image of every constructible subset is constructible it suffices to show that the inverse image of every retrocompact open V of Y is retrocompact in X, see Topology, Lemma 005I. The significance of V being retrocompact in Y is just that the open immersion $V \to Y$ is quasi-compact. Hence the base change $f^{-1}(V) = X \times_Y V \to X$ is quasi-compact too, see Schemes, Lemma 01K5. Hence we see $f^{-1}(V)$ is retrocompact in X. Suppose E is locally constructible in Y. Choose $x \in X$. Choose an affine neighbourhood V of $f(x)$ and an affine neighbourhood $U \subset X$ of x such that $f(U) \subset V$. Thus we think of $f|_U : U \to V$ as a morphism into V. By Properties, Lemma 054C we see that $E \cap V$ is constructible in V. By the constructible case we see that $(f|_U)^{-1}(E \cap V)$ is constructible in U. Since $(f|_U)^{-1}(E \cap V) = f^{-1}(E) \cap U$ we win. □